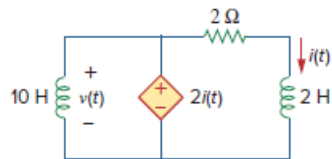


Problem

For the circuit in Fig. 16.73, find $v(t)$ for $t > 0$. Assume that $i(0) = 2$ A.

Figure 16.73:



Step-by-step solution

Step 1 of 2

Refer to circuit diagram in Figure 19.73 in the textbook.

The initial current through the 2 H inductor is 2 A and initial current through 10 H inductor is zero.

Draw the circuit in s-domain as shown in Figure 1.

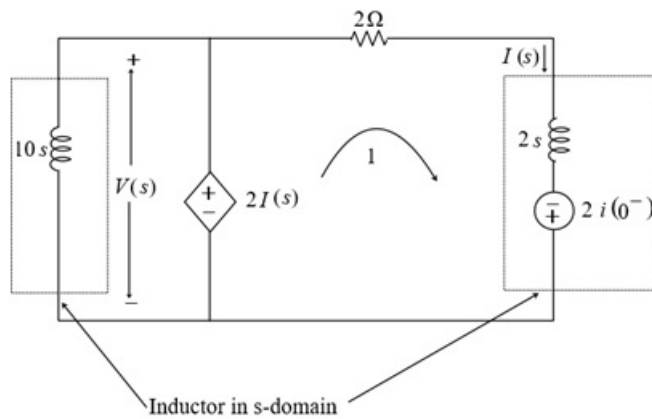


Figure 1

Step 2 of 2

Apply Kirchhoff's voltage law to loop 1.

$$-2I(s) + 2I(s) + 2sI(s) - 2i(0^-) = 0$$

$$2sI(s) = 2(2 \text{ A})$$

$$I(s) = \frac{2}{s}$$

Express $V(s)$ in terms of $I(s)$.

$$V(s) = 2I(s)$$

Substitute $\frac{2}{s}$ for $I(s)$.

$$\begin{aligned} V(s) &= \frac{2(2)}{s} \\ &= \frac{4}{s} \end{aligned}$$

Apply inverse Laplace transform.

$$\begin{aligned} v(t) &= \mathcal{L}^{-1}\left[\frac{4}{s}\right] \\ &= 4u(t) \text{ V} \end{aligned}$$

Therefore, the value of $v(t)$ is $\boxed{4u(t) \text{ V}}$.